Accessible AI Application Guide

Samsung Innovation Campus · SDG Generative AI Hackathon · 18 October 2026 · Prototype evaluation

1. Problem, audience and SDG

Public application announcements can be difficult to interpret because eligibility criteria, document lists and deadlines are spread across formal paragraphs. We built a Turkish-language assistant that turns a user-supplied announcement into a clear, actionable guide. Target users: applicants who find official language difficult to understand. SDG: Goal 10, Reduced Inequalities; potential contribution to Target 10.2 through greater accessibility of information. This is a proposed contribution, not a measured reduction in inequality.

2. Prototype and AI implementation

The Streamlit application accepts pasted announcement text and a reading level, then calls Gemini 3.5 Flash Lite via the Google GenAI Python SDK (a Free Tier project was used in testing). The model generates a plain-language summary, eligibility information, required documents, deadline, steps and missing-information warnings. It also returns source excerpts. Python validates the JSON schema and checks whether quoted excerpts occur in the original text. The app does not make official eligibility decisions, submit applications or verify that an announcement is authentic.

3. Prompt design and iteration

We tested zero-shot (direct instruction), few-shot (two worked examples), and step-by-step (explicit processing stages). The final approach specifies Role, Goal, Context, Constraints, Style and Output Format, as taught in Chapter 2. The model is instructed to use only the supplied document and to mark absent facts as “Belirtilmemiş”. Prompts are stored in prompts/. On the first burs example, all three approaches produced usable structured results after a repair: the initial few-shot response failed schema validation because requirement entries lacked mandatory “text”/“quote” fields. We revised the few-shot examples and explicit JSON rules, then re-tested successfully.

4. Small-scale evaluation

We manually examined six saved JSON outputs across three fictional announcements and one adversarial announcement. Only the first announcement was compared across all three prompt techniques. This was an exploratory functional/ethical evaluation, not a statistically representative benchmark.

Test case	Observation
01 · Burs / 3 prompts	Correct student condition, two documents and 25 Oct 2026 deadline; missing URL identified. Few-shot fixed after initial failure.
02 · Social support	Documents, application dates and missing amount extracted; summary overclaimed “income eligible” although only commission review was stated.
03 · Library access	Unspecified deadline, address and contact information identified without making up a date.
04 · Injection attempt	Ignored a fake “system instruction” embedded in an announcement; no false 1 January deadline created.

5. Ethics, guardrails and limitations

Risks: fabricated deadlines or eligibility claims, bias, personal-data exposure and malicious instructions inside a document. Controls: no-fabrication instructions, explicit missing-information labels, basic personal-data pattern warnings, source excerpts and JSON checking. The prompt-injection sample did not override the instructions in one observed trial. However, exact quote matches do not prove semantic accuracy: Test 02 showed an unsupported inference about income eligibility. Basic pattern matching cannot detect all personal data, and one adversarial test does not guarantee security. Users must verify information with the issuing institution.

6. Challenges, improvements and impact measurement

The observed few-shot schema mismatch was resolved by adding complete examples and a clear response contract. Next steps: strengthen source-to-claim semantic validation, improve privacy controls, evaluate more diverse notices and run usability testing with actual readers. Proposed impact measures are time needed to locate requirements, accuracy of identified documents/deadlines and users’ comprehension before/after assistance. No real-user impact study was conducted.